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REVIEW ARTICLE

Efficacy of imagery rehearsal therapy in the treatment of traumatic nightmares in youth with complex post-traumatic stress disorder

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Abstract

Background Traumatic nightmares are a core feature in adolescents with complex PTSD (CPTSD) exposed to early and repeated trauma, maintaining hyperactivity and dysregulation. Despite their prevalence and functional impact, targeted treatments for youth remain scarce. Imagery rehearsal therapy (IRT) is a brief cognitive-behavioral intervention effective in adults, but evidence in adolescents with CPTSD is limited.

Methods This study assessed the feasibility and preliminary effectiveness of four sessions of IRT adapted for young participants (14–25 years) with CPTSD. Thirty-nine participants presenting ICD-11 criteria for CPTSD were recruited after at least 3 months of psychiatric follow-up. They received IRT and completed pre- and post-intervention self-report measures of nightmare frequency and distress (NDQ-13), sleep quality (PSQI, ISI), PTSD symptoms (CPTS-RI, CPC, PCL-5), anxiety, depression, and quality of life.

Results All participants reported severe traumatic nightmares, mainly linked to repeated sexual (75%) and physical (65%) violence. After four IRT sessions, nightmare frequency per week decreased (5.7 vs. 2.8, $p=0.011$) as did the distress rate on the visual analogue scale between 0 and 10 (7.7 vs. 4.4, $p=0.004$). Sleep improved (PSQI 9.1 vs. 13.3, $p=0.007$; ISI 12.2 vs. 17.9, $p=0.005$), and quality of sleep (6.8 vs. 4.4, $p=0.01$) and of life increased (6.8 vs. 4.7, $p=0.014$), while anxiety and depressive symptomatology remained stable.

Conclusion IRT appears feasible and promising in this population and was associated with reduced nightmare frequency and distress, along with improved sleep and quality of life. These results highlight its potential as a safe and easily disseminable intervention, also suitable for vulnerable or low-resource populations.

Brief summary

Current knowledge/study rationale Traumatic nightmares are highly prevalent and impairing in adolescents with complex PTSD (CPTSD), yet developmentally adapted treatments remain scarce. Imagery Rehearsal Therapy is effective in adults, but evidence in youth exposed to early and repeated trauma is limited.

Study impact This study suggests that a brief, four-session IRT protocol is feasible and may reduce nightmare frequency and distress in adolescents and young adults with CPTSD. The findings support IRT as a safe and scalable intervention that may help improve sleep and functioning in vulnerable or low-resource populations.

Keywords Post-traumatic stress disorder · CPTSD · Traumatic nightmares · Sleep disturbances · Imagery rehearsal therapy · Adolescent · Youth

Introduction

PTSD in adolescents and young adults

Post-traumatic stress disorder (PTSD) affects around 15.9% of children one year after a traumatic event [1]. In adults, lifetime prevalence ranges from 1.3 to 12.2% [2]. Core

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symptoms include intrusive memories, avoidance, mood alterations, and hyperarousal [3]. Among youth, PTSD disrupts multiple developmental domains: attachment, emotional regulation, school adjustment [4, 5]. These early difficulties can persist into adulthood, severely impacting quality of life [6]. Complex PTSD (CPTSD), often linked to repeated or inescapable trauma, involves persistent disturbances in self-organization (emotional dysregulation, negative self-perception, relational difficulties) [7, 8]. Population studies estimate that 5.6% of adolescents meet ICD-11 criteria for CPTSD versus 2.3% for simple PTSD [9], with 3.8% prevalence in adults [10].

Sleep disturbances in trauma-exposed youth

Sleep disturbances are an early, sensitive, and persistent marker of PTSD in children and adolescents. They are recognized both as symptoms and as risk and maintenance factors for PTSD [11, 12]. Difficulties falling asleep, related to hyperactivation or behavioral avoidance, often worsen after trauma [13]. Sleep quality is typically impaired, with frequent awakenings and increased daytime sleepiness [14]. Sleep architecture is disrupted, with reduced non-REM consolidation and fragmented REM sleep [15]. Critically, traumatic nightmares are a core PTSD symptom, marked by high recurrence (≥ 2/week), intense sensory content, and strong autonomic activation [16]. These nightmares often replay the traumatic scene, fueling distress and avoidance. Other parasomnias (night terrors, sleepwalking) may also emerge after trauma [17].

Traumatic nightmares: features and impact

Traumatic nightmares are stereotyped, repetitive, and highly sensory (visual, tactile, olfactory) [18]. They provoke abrupt awakenings with screaming or crying, leading to hypervigilance and bedtime avoidance. Unlike ordinary nightmares, which are more varied, narrative, and less emotionally charged, traumatic nightmares are consistent and clinically significant [19]. Beyond being distressing, they reinforce trauma-related memory networks, maintaining the vividness and emotional charge of the memory [20–22]. Repeated nocturnal re-experiencing disrupts emotional processing and memory consolidation, creating a vicious cycle: hyperarousal, REM sleep fragmentation, and sleep debt worsen daytime emotional reactivity and reduce regulation [23, 24]. Thus, traumatic nightmares are not mere symptoms but major pathogenic mechanisms of uncontrollable nocturnal re-experiencing that sustain and worsen PTSD [25, 26]. Their chronicity may also compromise first-line PTSD treatment effectiveness [27, 28], while targeted sleep interventions can improve recovery [29]. Imagery rehearsal therapy (IRT) is the reference specific treatment for nightmares, trauma-related or not.

Imagery rehearsal therapy (IRT) in youth with PTSD and CPTSD

IRT, developed in the 1990 s by Barry Krakow and colleagues, aims to reduce nightmares' impact through cognitive restructuring and positive imagery [30]. Based on dream continuity theory, it involves replacing the nightmare with a controllable, reassuring scenario. Patients' practice visualizing this new content to reduce the original nightmare's emotional impact. IRT is recommended as first line by the American Academy of Sleep Medicine. Its efficacy is well demonstrated in adults, reducing nightmare frequency and distress and improving sleep [31, 32]. However, most studies focus on adults with "classic" PTSD, with limited research in adolescents or in CPTSD populations [31, 33, 34]. Given this population's developmental and emotional specificities, evaluating IRT in youth is essential.

Study objectives

Our study aims to assess the feasibility and preliminary efficacy of IRT in reducing traumatic nightmares, improving sleep, circadian rhythms, and alleviating PTSD/CPTSD symptoms in adolescents and young adults. Secondary objectives include evaluating effects on comorbidities (anxiety, depression, emotional regulation) and quality of life, as well as improving the clinical characterization of traumatic nightmares to further adapt IRT protocols for this population.

Materials and methods

Participants and study setting

Adolescents and young adults aged 14 to 25 years were recruited between August 2022 and January 2025 at the Sleep Disorders Center/International Center for Research in ChronoSomnology (CIRCSom) at Strasbourg University Hospitals. Inclusion criteria comprised a diagnosis of complex PTSD (CPTSD) according to ICD-11 criteria [35]. In practice, CPTSD was established by experienced clinicians during routine psychiatric assessment, based on the presence of core PTSD symptoms together with persistent disturbances in self-organization (DSO), including affective dysregulation, negative self-concept, and relational difficulties. This diagnostic formulation was supported by the International Trauma Questionnaire (ITQ), which was used to assess ICD-11 PTSD and

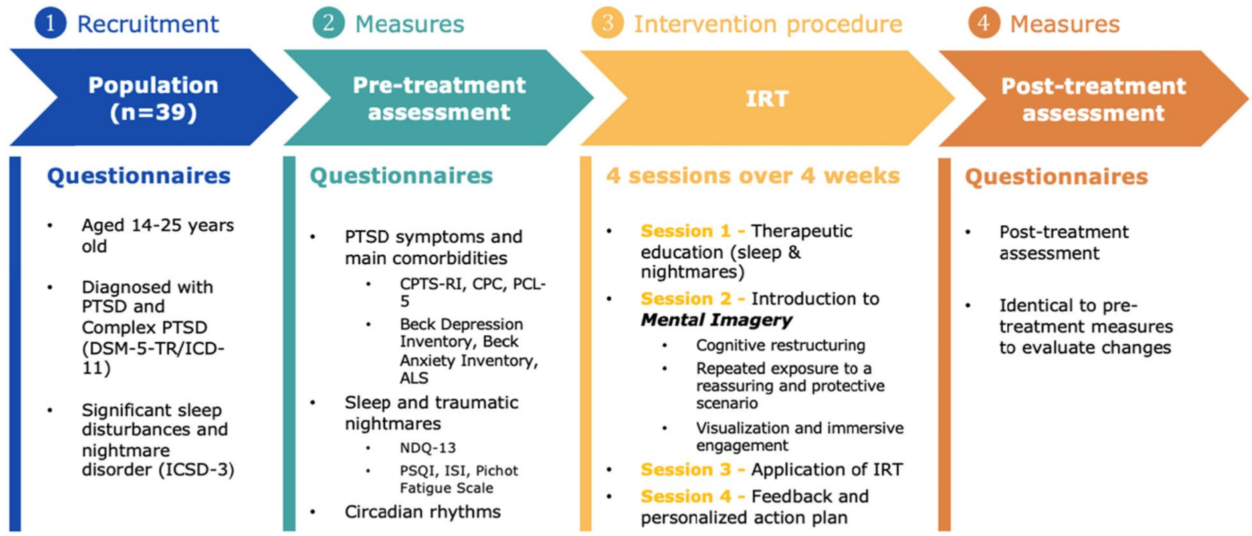


Fig. 1 Study design and Imagery Rehearsal Therapy procedure. *ALS*, Affective Lability Scale; *CPC*, Child PTSD Checklist; *CPTS-RI*, Child Posttraumatic Stress Reaction Index; *DSM-5*, Diagnostic and Statistical Manual of Mental Disorders, 5th Edition; *ICD-11*, International Classification of Diseases, 11th Revision; *IRT*, imagery rehearsal therapy; *ISI*, Insomnia Severity Index; *NDQ*, Nightmare Distress Questionnaire; *PCL-5*, PTSD Checklist for DSM-5; *PSQI*, Pittsburgh Sleep Quality Index

CPTSD symptom profiles. The ITQ was therefore used as a structured support to clinical evaluation rather than as a standalone diagnostic instrument. Participants also had to meet ICSD-3 criteria for nightmare disorder (recurrent, distressing REM sleep nightmares with clear recall and daytime impact) [36].

Adolescents were required to assent to participation, have obtained parental consent, and be engaged in at least three months of child psychiatric follow-up. Exclusion criteria included severe developmental delay, psychotic disorders, participation in other studies evaluating psychotherapies or pharmacological treatments for PTSD-related sleep disorders, or the presence of “content-free nightmares” (abrupt awakenings with strong bodily and emotional activation but no remembered dream content), which accounted for 25% of traumatic nightmare cases treated during this period in our department.

Intervention procedure

All 39 participants (see Fig. 1) received four individual sessions of imagery rehearsal therapy (IRT), following the main steps of the protocol developed by Barry Krakow and colleagues (1988, 1999): (i) psychoeducation and cognitive restructuring, (ii) nightmare selection and modification, and (iii) mental imagery rehearsal, delivered as weekly 90-min sessions.

The content and duration of sessions were adapted to developmental level and trauma complexity, with additional psychoeducation components targeting trauma (see Table 1). Therapy was delivered by psychiatrists specialized in sleep and psychotraumatology at Strasbourg University Hospitals, in collaboration with the Child and Adolescent Psychiatry Department, as part of routine clinical care. All therapists involved in the study were psychiatrists working in sleep medicine and psychotraumatology and were trained in the same Imagery Rehearsal Therapy (IRT) module delivered by Dr. Agnès Brion (Paris), a senior psychiatrist with expertise in sleep medicine and psychotraumatology and an experienced continuing education trainer in IRT. The intervention followed a shared standardized four-session protocol derived from Krakow’s model and adapted by the authors for adolescents and young adults with CPTSD. To strengthen reproducibility, therapists used the same session structure, core therapeutic targets, and psychoeducational framework. Limited flexibility was allowed only for developmentally appropriate adaptations, modulation of emotional intensity, and the possible inclusion of a parent or reassuring attachment figure when clinically indicated. In addition, cases were discussed regularly in team supervision meetings in order to support consistency of delivery across therapists.

Measures

Sociodemographic and clinical characteristics

Age, sex, and place of residence were extracted from medical-administrative records completed at admission. Clinical interviews assessed trauma exposure (type, date, frequency), family situation, protection measures, and psychiatric care [35–37].

Table 1 Imagery rehearsal therapy intervention procedure and main adjustments for adolescents with PTSD

Imagery rehearsal therapy intervention procedure*
Session 1 – Sleep and traumatic nightmares therapeutic education (90 min)
o Therapeutic education on sleep physiology and its functions in a psychotraumatology context
o Definition of dreams and introduction to the continuity hypothesis (impact of daily experiences on dreams)
o Explanation of nightmares (causes, link with psychotrauma, impact on sleep and wakefulness)
o Reflection on the participant's relationship with their nightmares and their level of adherence to them
Session 2 – Introduction to positive mental imagery (90 min)
o Explanation of the automatic nature and repetitive patterns of nightmares
o Presentation of mental imagery as a cognitive restructuring tool
o Practical exercises in positive mental imagery (visualizing a pleasant memory or a safe scene)
Session 3 – Application of IRT (90 min)
o Review of the participant's experience with mental imagery between sessions
o Implementation of IRT: active modification of the target nightmare scenario (if multiple nightmares, prioritize a moderately distressing one)
o Guided exercises to transform distressing dreams (modifying the traumatic scenario or imagining an alternative scenario)
o Practical application: IRT focused on a scenario for approximately 10 min per day
Session 4 – Feedback and personalized action plan (90 min)
o Discussion on progress made and difficulties encountered
o Planning for autonomous practice of IRT while anticipating potential obstacles
o Focus on practicing a maximum of one scenario over the following two weeks
Main adjustments for adolescents and young adults with PTSD
1. **Motivational work**
o By reminding them of the concrete goal of the IRT ("to be less afraid at night")
2. **Modulating the emotional load of sessions**
o By keeping sessions short and structured, and introducing reassuring opening and closing rituals right from the start
3. **The association of a parent or a reassuring mentor when indicated**
o By integrating this reassuring figure into the new dream scenario
4. **Active support for the construction of the alternative scenario**
o By using supports (e.g., cards, photo-language)
o To stimulate imagination, free association and the senses, while reducing verbal pressure
5. **Incorporating flexible imagery with intermediate stages**
o e.g., by first modifying a detail of the nightmare before proposing a complete new dream, or by concealing a nightmare
6. **Encourage playful repetition and motivation through challenge**
o By transforming the mental rehearsal exercise into a game or personal challenge (e.g., "succeed in adding a new positive element every day")

*Protocol adapted to youth population by the authors from Krakow et al. [30]

Assessment of PTSD symptoms

The effectiveness of IRT on sleep disorders, traumatic nightmares, PTSD symptoms, and comorbidities was measured via self-report at baseline (T0, 2 weeks before therapy) and post-treatment (T1, 6 weeks after the end of the 4 IRT sessions) (see Fig. 1). PTSD symptoms were assessed using:

- For participants aged 14–18:
    - o Child PTSD Checklist (approved for ages 14–18) (CPC; 40 items for 14–18 years; 23 symptom items, 6 functional impairment items; cut-offs ≥ 23 for PTSD symptoms, ≥ 4 for impairment) [38].
    - o Child PostTraumatic Stress Reaction Index (approved for ages 6–17) (CPTS-RI; 20 items, cut-offs: 12–24 mild, 25–39 moderate, 40–59 severe, ≥ 60 very severe; Cronbach alpha = 0.87 for French version), which was added as a supplement to clarify the severity of the symptoms [39].
- For participants aged 18 and over:
    - o PTSD Checklist for DSM-5 (approved for ages 18 and over) (PCL-5; 20 items for ages 18–25), scored 0–4; total score ≥ 32 suggests probable PTSD [40].

In addition to age-specific PTSD symptom scales, the International Trauma Questionnaire (ITQ [41]) was used to support the clinical characterization of ICD-11 CPTSD, particularly the assessment of DSO dimensions.

Subjective assessment of sleep and traumatic nightmares

Sleep and circadian rhythms were assessed with self-report questionnaires and a 15-day home sleep diary.

- Insomnia Severity Index (ISI; 0–7 no insomnia, 8–14 subclinical, 15–21 moderate, 22–28 severe) [42].
- Pittsburgh Sleep Quality Index (PSQI; score 0–21, higher = poorer sleep quality) [43].
- Pichot Fatigue Scale (> 22/36 indicate excessive fatigue).
- Nightmare Distress Questionnaire (NDQ; 13 items, score > 20 indicates significant distress) [44], with added custom items on post-traumatic parasomnias and visual analog scales for nightmare-related distress. Before the intervention, we collected detailed nightmare content in the sleep/dream diary (see circadian rhythms section) using a standardized instruction limited to a maximum of 15 lines, allowing participants to freely describe their traumatic nightmares (instructions: "Describe your nightmare concisely in a few lines [5–10], including the elements that are important to you"). These narratives were analyzed thematically by two independent raters using a predefined clinically informed coding grid (e.g., repetition of the traumatic scene, loss of control, inability to escape, threatening figures, absence of protection, symbolic distortions, and narrative disorganization). Disagreements were resolved by consensus. As this analysis was exploratory and descriptive, reported frequencies should be interpreted as indicative of recurring themes rather than precise quantitative estimates.

Subjective assessment of circadian rhythms

Chronotype [45] was assessed with:

- Horne & Ostberg questionnaire (H&O) [46, 47]; scoring: ≥ 70 = extreme morning type, ≤ 30 = extreme evening type.
- Participants completed sleep/dream diaries pre- and post-therapy (T0, T1), noting bedtimes, sleep latency, waking times, naps, dream frequency, and content.

Assessment of PTSD comorbidities

IRT effects (T0–T1 difference) on comorbidities were assessed via:

- Beck Depression Inventory (21 items; 0–9 minor, 10–18 mild, 19–29 moderate, 30–63 severe) [48].
- Beck Anxiety Inventory (21 items; 0–7 minor, 8–15 mild, 16–25 moderate, 26–63 severe) [49].
- Short Affective Lability Scale (ALS; > 21/54 indicates emotional lability) [50].

Procedure, legal, and ethical framework

The study was approved by the University of Strasbourg Ethics Committee (CE-2023–85). All data were collected using standardized and validated assessment instruments, in accordance with routine follow-up procedures at Strasbourg University Hospitals. Adherence to treatment (IRT completion rate) and feedback from professionals were also included. Participant IRT acceptability and satisfaction was assessed using visual analog scales (continuous variables ranging from 0—dissatisfaction to 10—very satisfied).

Statistical analysis

All analyses were performed using Jamovi software (Version 0.9.5.12). Non-parametric analyses were carried out in view of the small number of subjects ($n = 39$). First, we analyzed the data collected by questionnaire according to traumatological, sleep and circadian rhythm characteristics. Qualitative data were described in terms of numbers and percentages, while quantitative data were described in terms of averages and distributions. We also performed non-parametric paired analyses (Wilcoxon signed-rank test) to compare results before and after IRT psychotherapy. In addition to p-values, effect sizes were calculated for the main Wilcoxon signed-rank tests to provide a more clinically informative interpretation of pre-post changes. No formal correction for multiple comparisons was applied, as this study was exploratory in nature. We used a significance level of 5% corresponding to $p < 0.05$ ($^*p < 0.05$).

Results

Our study population consisted of 39 patients with post-traumatic nightmares associated with PTSD, including 32 girls and 7 boys aged between 14 and 25 years (34 minors and 5 adults) (see Table 2). Analysis of traumatic exposure revealed that these events mainly occurred during childhood (95%), with predominant sexual (75%) and physical (65%) abuse. All participants had experienced repeated trauma (100%) according to clinical assessment, with 46% exposed to cumulative or simultaneous traumatic events contributing to CPTSD.

Clinical severity was high across the sample (see Table 3). The mean CPTS-RI score was 54.7, with 66% classified as severe and 17% as very severe PTSD. The diagnostic CPC had a mean score of 49.9, and the PCL-5 had a mean score of 58.4, indicating major functional impact for 83% of minors (CPC functional impact). Before IRT, 100%

Table 2 Sociodemographic characteristics and traumatic exposure of the sample

Demographic variables	Results
N	39
Gender, *N* (%)	
Female	32 (82%)
Male	7 (18%)
Age (years)	
Range	14–25
Mean (standard deviation)	16.4 (3.1)
Type of complex trauma	
Sexual violence (e.g., rape)	29 (75%)
Physical violence	25 (65%)
Civilian war trauma	14 (35%)
Accidents	13 (33%)
Direct witness of traumatic events (e.g., murder)	12 (30%)
Miscarriage	1 (2.6%)
Cumulative and complex traumatic exposure	
Repeated traumatic exposure	39 (100%)
Co-occurrence of multiple types of violence	33 (85%)
Age of onset of abuse	
Before age 18	37 (95%)
After 18 years	2 (5%)
Child protection measure	
History of challenging out-of-home placement	18 (45%)

Data are presented in number (and percent, %), age is presented in years

met PTSD diagnostic criteria (CPC > 20, PCL 5 > 33), confirming pervasive and clinically significant post-traumatic symptoms.

Nightmares were prominent for all participants, with a mean NDQ-13 score of 37 exceeding the clinical threshold in 100% of cases. Participants described highly anxiety-provoking content directly linked to traumatic events, with significant functional repercussions: 100% reported recalling nightmares upon awakening, 92% reported awakenings due to nightmares, bedtime anxiety in 92%, and bedtime avoidance in 75%, and 85% experienced intrusive memories during the day.

Phenomenological descriptive thematic analysis of participants' narratives showed that traumatic nightmares were most often directly linked to the traumatic experience and were consistently described as highly anxiety-provoking. The most recurrent themes were helplessness, loss of control, inability to escape, passive exposure to threat, violence, and the absence of protection or reassuring figures (see Fig. 2). Many narratives also displayed symbolic distortions and narrative disorganization.

Sleep disturbances and fatigue were prominent across the sample. The mean PSQI score was 13.3, confirming poor sleep quality (see Table 3), with 85% of participants reporting poor sleep overall. Insomnia was moderate to severe in 77–85% of cases, while excessive daytime sleepiness was observed in 46% (mean Pichot Sleepiness Scale score of 21.7). Subjective sleep quality was rated at 4.4/10, and nightmare-related anxiety at 7.7/10. In addition, 69% of participants exhibited nocturnal behavioral manifestations such as motor agitation or vocalizations. Circadian profile analysis revealed 15% morning-type, 62% neutral, and 23% evening-type chronotypes. Moreover, 85% reported post-traumatic changes in sleep patterns, particularly delayed bedtimes, highlighting the pervasive impact of trauma on sleep–wake rhythms, and sleep debt, with 46% of participants reporting excessive levels. Comorbidities were common, with Beck scores indicating severe anxiety and depressive symptomatology (34.6), with 77% and 84% respectively exceeding clinical thresholds. The ALS confirmed emotional instability in 83% of participants (mean score: 32.3). Quality of life was rated at 4.4/10.

Moreover, CPTSD was significantly correlated to comorbidities and sleep. The NDQ-13 score correlated with nightmare anxiety on the VAS ($r = 0.664$, $p = 0.01$). The CPTS-RI scale correlated with anxiety (Beck) ($r = 0.899$, $p = 0.01$), as did functional impact (CPC) ($r = 0.928$, $p < 0.01$). In addition, strong correlations were found between depressive symptomatology (Beck) and CPC ($r = 0.943$, $p = 0.01$), CPTS-RI ($r = 0.928$, $p < 0.01$) and PCL-5 ($r = 0.821$, $p = 0.03$) scores, suggesting a robust association between PTSD and anxious and depressive symptomatology, in line with the literature. Sleep quality was positively associated with quality of life ($r = 0.581$, $p < 0.01$).

After IRT, PTSD symptom scores decreased in the adult subgroup on the PCL-5, although this result must be interpreted with caution given the very small number of adults ($n = 5$). In adolescents, PTSD symptom measures showed a non-significant trend toward improvement (Fig. 3 and Table 3). Nightmares were also significantly less frequent and less distressing, with a reduction in the NDQ-13 total score ($p = 0.011$) and in the "fear" subscale ($p = 0.012$). Sleep quality improved significantly, as evidenced by significant decreases in the PSQI ($p = 0.007$) and ISI ($p = 0.005$) scores. Perceived quality of sleep ($p = 0.01$) and quality of life ($p = 0.014$) increased, while anxiety related to nightmares decreased significantly ($p = 0.004$). No statistically significant changes were observed for anxiety, depressive symptoms, fatigue, or affective lability after IRT, although descriptive score reductions were noted for some of these measures.

These findings suggest a potential benefit of IRT on traumatic nightmares and sleep-related outcomes. Finally, acceptability as measured by a visual analog scale was 100%, with a satisfaction rate of 95%.

Table 3 Imagery rehearsal therapy treatment outcome measures

	Pretreatment T0			Posttreatment			
Scale	N	T0 mean	SD	T1 mean	SD	***p***	Effect size (***r***)
CPC total	34	60.84	20.59	52.5	16.2	0.203	0.22
CPC diagnostic PTSD	34	49.92	13.52	42.1	11.7	0.106	0.28
CPC functional impact	34	11.02	7.18	10.3	5.6	0.832	0.130
CPTS-RI total	34	54.7	10.4	51.3	10.3	0.844	0.128
PCL-5 total	5	58.34	13.1	44.9	21.1	0.034*	0.832
NDQ-13 total score	39	35.92	8.15	29.0	10.58	0.011*	0.749
NDQ-13 fear	39	15.31	4.80	12.38	5.62	0.012*	0.741
NDQ-13 interference	39	13.00	2.35	10.54	4.31	0.054	0.594
NDQ-13 premonition	39	4.62	1.39	3.69	1.55	0.113	0.492
PSQI total	39	12.85	3.26	9.08	3.52	0.007*	0.785
Pichot fatigue scale total	39	20.92	6.50	16.0	9.38	0.084	0.510
ISI total	39	17.23	2.68	12.23	5.05	0.005*	0.816
Horne & Ostberg	39	45.54	11.06	48.08	10.20	0.169	0.408
VAS sleep quality (/10)	39	4.67	1.37	6.57	2.28	0.010*	0.759
VAS quality of life (/10)	39	4.68	1.58	6.58	2.63	0.014*	0.689
VAS nightmare-related anxiety (/10)	39	7.38	2.32	4.43	3.35	0.004*	0.885
Beck anxiety inventory total	39	28.77	12.71	23.62	13.19	0.069	0.515
Beck depression inventory total	39	34.77	13.74	29.54	16.95	0.076	0.523
ALS total	39	32.33	14.81	25.67	14.80	0.279	0.17

Pretreatment (T0) and posttreatment (T1) outcomes after imagery rehearsal therapy. Data are shown as mean ± standard deviation. P-values were obtained using Wilcoxon signed-rank tests. Corresponding effect sizes are reported as r values. *ALS*, Affective Lability Scale; *CPC*, Child PTSD Checklist; *CPTS-RI*, Child Post-Traumatic Stress Reaction Index; *ISI*, Insomnia Severity Index; *NDQ-13*, Nightmare Distress Questionnaire-13; *PCL-5*, PTSD Checklist for DSM-5; *PSQI*, Pittsburgh Sleep Quality Index; *PTSD*, post-traumatic stress disorder; *VAS*, Visual Analog Scale

Discussion

Clinical and demographic characteristics of participants

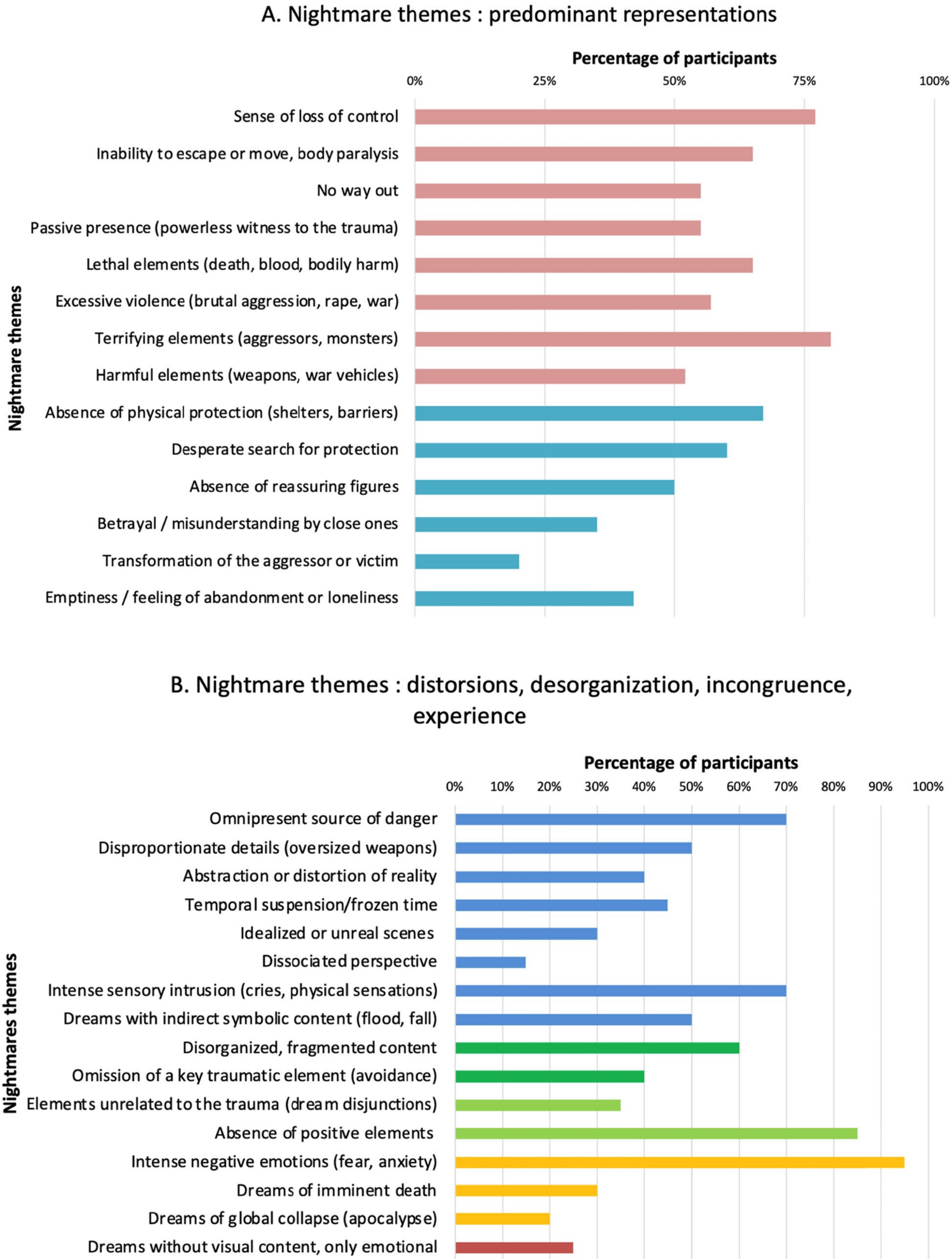
Our results highlight the severity of sleep disturbances, particularly nightmares and insomnia, among youth with CPTSD secondary to early and repeated childhood trauma. These disturbances significantly impair sleep quality [11], as reflected in high scores on the PSQI, NDQ-13 (nightmares), and ISI (insomnia) scales [51], consistent with the work of Germain et al. [22] and Spoormaker & Montgomery [52]. The high prevalence of depressive symptomatology, the most common PTSD comorbidity, appears to be exacerbated by the chronicity of posttraumatic symptoms and associated sleep disturbances [12, 29]. With regards to affective lability, a core feature of CPTSD, this is evidenced by an ALS-18 score of 33.5, indicating marked emotional instability at a level intermediate between the general population and scores described for severe emotion regulation disorders.

Our sample also showed a female overrepresentation (82%), well documented in the literature [54], and explained by various authors by greater exposure to interpersonal violence, particularly sexual violence [53], a higher tendency to verbalize emotional distress, and recruitment biases associated with help-seeking behaviors. Some authors also suggest an overall lower level of resilience in females [54].

Phenomenology of traumatic nightmares before treatment

Qualitative analysis of reported nightmares confirmed classically observed phenotypic criteria: stereotyped and repetitive content, heightened sensory realism (visual, auditory, olfactory, tactile), poor narrative structure, and fragmented content [26, 55]. These features, also observed among war-exposed youth (e.g., Punamäki's work with Kurdish adolescents), indicate a profound alteration in dream processing, characterized by impoverished and disorganized narrative sequences [56].

Most participants described nightmares directly linked to traumatic events, with intense emotional content and abrupt awakenings in over 90% of cases [57, 58]. The absence of dream-like distancing, with the experience perceived as "raw reality" [55], reinforces the perception of constant threat and contributes to significant daytime consequences. Persistent recall of the dream upon waking (100%), daytime intrusions (85%), and anticipatory anxiety


A. Nightmare themes : predominant representations
Percentage of participants
0%
25%
50%
75%
100%
Nightmare themes
Sense of loss of control
Inability to escape or move, body paralysis
No way out
Passive presence (powerless witness to the trauma)
Lethal elements (death, blood, bodily harm)
Excessive violence (brutal aggression, rape, war)
Terrifying elements (aggressors, monsters)
Harmful elements (weapons, war vehicles)
Absence of physical protection (shelters, barriers)
Desperate search for protection
Absence of reassuring figures
Betrayal / misunderstanding by close ones
Transformation of the aggressor or victim
Emptiness / feeling of abandonment or loneliness
B. Nightmare themes : distorsions, desorganization, incongruence, experience
Percentage of participants
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
Nightmares themes
Omnipresent source of danger
Disproportionate details (oversized weapons)
Abstraction or distortion of reality
Temporal suspension/frozen time
Idealized or unreal scenes
Dissociated perspective
Intense sensory intrusion (cries, physical sensations)
Dreams with indirect symbolic content (flood, fall)
Disorganized, fragmented content
Omission of a key traumatic element (avoidance)
Elements unrelated to the trauma (dream disjunctions)
Absence of positive elements
Intense negative emotions (fear, anxiety)
Dreams of imminent death
Dreams of global collapse (apocalypse)
Dreams without visual content, only emotional

◂**Fig. 2** Nightmare themes. **A** Predominant representations. **B** Distorsions, desorganization, incongruence, experience. Recurrent nightmare themes identified in participants' narratives. Bars indicate descriptive frequencies, with no inferential or psychometric value. **A** Presents predominant representation (loss of control, inability to escape, threatening figures, violence, absence of protection). **B** Presents distortions, disorganization, incongruence, and subjective experiences. Multiple themes could be coded within the same narrative

(92%) promote sleep-avoidance behaviors (75%), leading to severe deterioration in sleep quality and the development of chronic hyperarousal and insomnia. Additionally, nightmares were frequently associated with parasomnias (69%), mainly confusional arousals and hypnagogic hallucinations, reflecting disrupted sleep architecture. Unlike night terrors, confusional arousals may persist during adolescence, especially in contexts of stress or sleep debt, and seem to be increasing in this age group. In this context, PTSD, with its physiological hyperactivation and dissociative aspects, promotes the emergence of "borderline" parasomnias [59], exacerbated by REM sleep fragmentation and reduced slow-wave sleep [60].

Some authors suggest that such combinations of nightmares, confusional arousals, and abnormal motor behaviors may represent a distinct sleep disorder—"Trauma-Associated Sleep Disorder" (TASD)—warranting specific diagnostic and therapeutic approaches [59].

Traumatic themes and emotional content of nightmares

In terms of content, nightmares were marked by themes of violence, insecurity, and powerlessness: absence of protective figures (85%), scenes of aggression or threat (80%), lack of safety (67.5%), and death-related experiences (65%). These themes are consistent with those observed in studies of war veterans and children exposed to armed conflict (Harb et al. on children from Gaza; Punamäki et al. on children from Kurdistan), reflecting loss of control and helplessness often amplified as paralysis, passive witnessing, or futile escape attempts [26, 61–63].

Nightmare-related emotional distress before treatment was particularly severe (NDQ-13 mean: 37.3), exceeding values reported in adults with PTSD [64]. This severity can be explained by the cumulative and chronic nature of the trauma experienced [65, 66], emotion regulation difficulties typical of adolescence, and recruitment from specialized hospital units in psychotraumatology and sleep. Moreover, the intense emotional content of participants' nightmares (95%) suggests material that is difficult to process, serving as both a marker of unresolved psychological distress and of impaired nocturnal emotional regulation [22, 56, 67, 68]. According to Hartmann, nightmares are not simply replays of traumatic events but emotionally charged dream productions attempting to integrate intense affects—fear, vulnerability—into memory processing [69, 70]. In this regard, 65% of the nightmares in our sample involved literal repetition of the traumatic scene, suggesting inefficient memory retrieval processes [71]. These encoding and consolidation dysfunctions [72, 73] may explain narrative fragmentation (60%), temporal distortions, and constant feeling of threat (70%) [56, 63].

When placing our qualitative findings within the nightmare severity spectrum proposed by Mysliwiec et al., qualitative items were grouped by correspondence to the four described categories, and mean percentages were calculated for each type [59]. It appears that 100% of participants reported nightmares directly linked to their traumatic experience. Among them, 65% corresponded to replicative nightmares, characterized by the literal reproduction of the traumatic scene (see Fig. 4), with a mean severity profile of about 64% across the different clinical markers. Non-replicative nightmares, marked by symbolic distortions or fragmented narratives, were observed in approximately 60–70% of the youth, corresponding to an average prevalence of 56%. In contrast, more incongruent or symbolic elements, consistent with posttraumatic dreams, were present in about 35% of narratives, with representation of traumatic exposure for 40% of them. The broad category of posttraumatic nightmares, encompassing the core clinical features common across subtypes, showed the highest overall profile with an average of 86%, underlining their centrality in post-traumatic dream life.

In addition, about 25% of the children assessed at the regional psychotrauma center reported nocturnal awakenings following trauma exposure without dream recall yet experienced as equivalent to a nightmare on both the emotional and physiological level. Although excluded from our main analyses, their frequency and clinical characteristics suggest that they represent a distinct form within the spectrum of traumatic nightmares, nocturnal panic attacks, or night terrors. We propose designating these phenomena as amnesic nightmares, hypothesizing that they occur in a posttraumatic context marked by intense physiological activation and dissociative mechanisms that prevent access to dream content. They may reflect trauma exposures leading to strong dissociation that blocks access to dream imagery, or early developmental trauma occurring at a stage when symbolic representational capacities were still immature.

This proposal is consistent with recent approaches encouraging a shift beyond dream content-based analyses to focus more on the repetitive, emotional, and physiological dimensions of traumatic nightmares [21, 74]. Integrating amnesic nightmares into the existing typology would thus provide a more comprehensive account of the diversity of clinical forms and deepen our understanding of the underlying psychopathological processes,

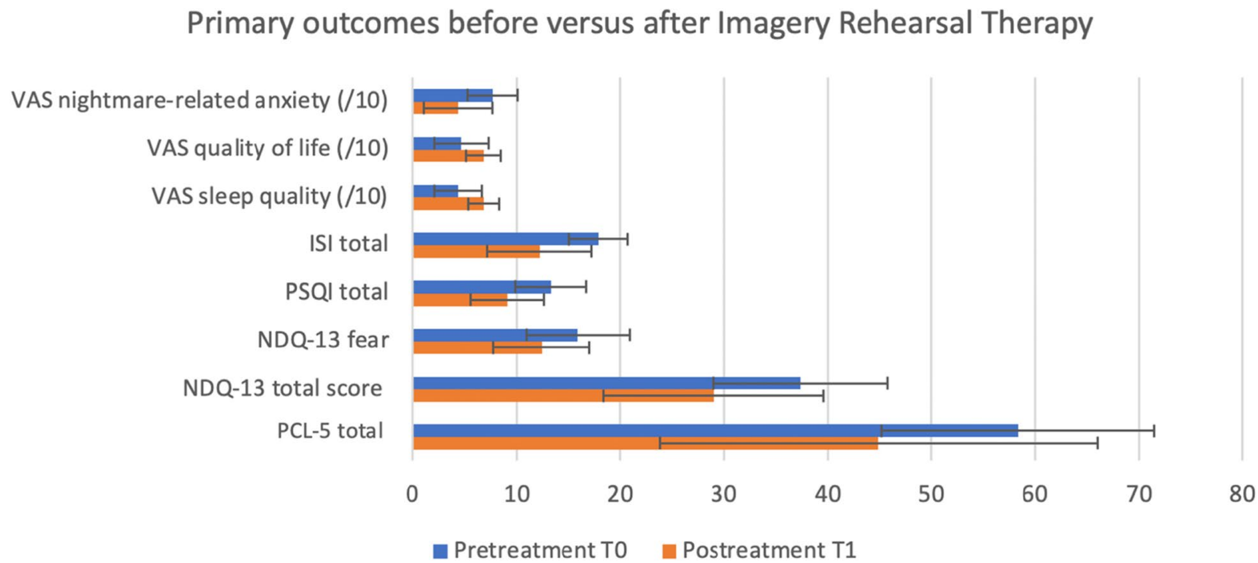


Fig. 3 Primary outcomes before and after imagery rehearsal therapy. Primary outcomes at baseline (T0) and posttreatment (T1). Bars show means and error bars standard deviations. Measures included PCL-5 (adults only, $n=5$), NDQ-13, PSQI, ISI, and VAS ratings for sleep quality, quality of life, and nightmare-related anxiety (all others, $n=39$ unless otherwise specified). Higher scores reflect greater symptom severity, except for sleep quality and quality of life. $p<0.05$ for paired pre-post comparisons. *ISI*, Insomnia Severity Index; *NDQ-13*, 13-item Nightmare Distress Questionnaire; *PCL-5*, PTSD Checklist for DSM-5; *PSQI*, Pittsburgh Sleep Quality Index; *T0*, pretreatment; *T1*, posttreatment; *VAS*, visual analog scale

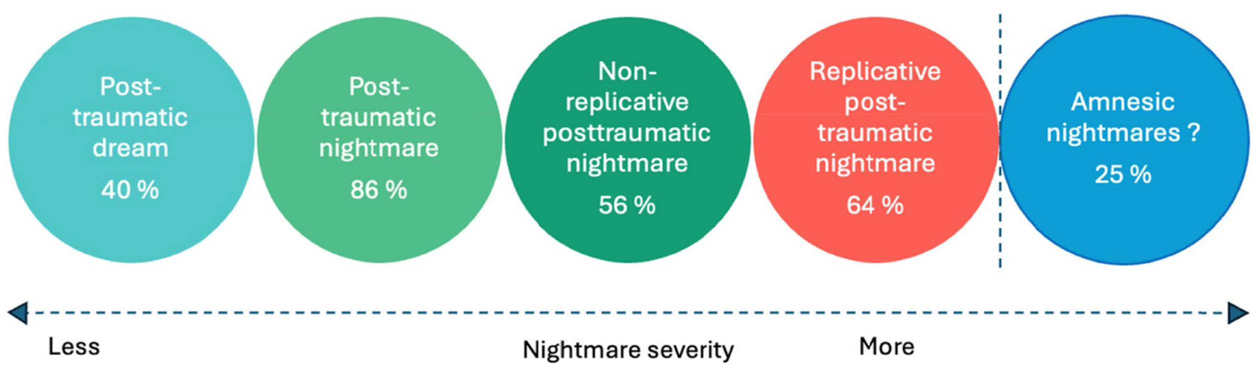


Fig. 4 Spectrum of trauma-related nightmares, with proposed integration of amnesic nightmares (reproduced and adapted from Mysliwiec et al. [59]). Model illustrating a continuum of severity in post-traumatic nightmares, from symbolic dreams to replicative nightmares reflecting true nocturnal re-experiencing. Amnesic nightmares are added as awakenings with intense emotional and physiological activation but no dream recall. Adapted from Mysliwiec et al. [59]

particularly in children and adolescents. Importantly, these categories are not mutually exclusive: the same child may report different types of nightmares depending on the nature of the traumatic exposure and the night, illustrating the variability and complexity of post-traumatic dream life.

The clinical features of participants' nightmares support the hypothesis that sleep disturbances contribute to the maintenance of hyperarousal and the failure of fear extinction [22, 25, 26, 52, 57], sustaining a vicious cycle between daytime symptoms and nightmares. IRT aims to break this cycle by promoting more structured and positive dream content, enabling more adaptive cognitive-emotional processing of trauma [24, 75].

Efficacy of imagery rehearsal therapy

After IRT, nightmares were associated with being less frequent and distressing (NDQ-13 total: $p=0.011$; "fear" subscale: $p=0.012$) [76–79], with significant improvement in subjective sleep quality (PSQI: $13.3 \rightarrow 9.1$; $p=0.007$ vs. $13.6 \rightarrow 10.0$; $p=0.002$)) [61, 62, 77, 78, 80]. These findings suggest a potential benefit of IRT on nightmare-related distress and sleep quality in this population, although they cannot be interpreted as proof of efficacy given the absence of a control condition. In addition, these effects have been described as particularly strong when dream rewriting addresses the central traumatic theme [61]. The significant reduction in nightmare-related emotional distress aligns with Krakow [81] and recent studies [62, 78, 82], including Sayk et al. [83], conducted in adults with borderline personality disorder (BPD), over 90% of whom had experienced repeated and severe trauma. This strong comorbidity and shared transdiagnostic vulnerabilities between CPTSD and BPD support exploring developmental and symptomatic continuity, as well as the possible relevance and adaptability of IRT in these clinical populations.

In contrast, the partial reduction in anxiety and PTSD symptoms (PCL-5: 58.3 → 44.9; $p = 0.034$ vs. IES-R intrusions: 25.92 → 19.38; hyperarousal: 22.62 → 14.77) was observed only in adults, not adolescents. These PTSD-related findings should be interpreted cautiously, as adolescents and young adults were assessed with different age-adapted instruments and the adult subgroup was very small. In addition, developmental differences in imagery abilities, cognitive maturation, and emotional regulation may influence responsiveness to IRT. Accordingly, changes observed on the PCL-5 should be viewed as preliminary and descriptive rather than as robust evidence of age-specific treatment effects. This pattern may be consistent with previous literature [62, 76, 81] and may reflect adolescents' emotional and cognitive immaturity, which can limit the impact of targeted interventions such as IRT. While IRT may help reduce nightmares, it does not directly address other core PTSD symptoms like hypervigilance, flashbacks, or avoidance, which often require more integrative approaches. Among adults, results are mixed: some studies report global symptom improvement [77, 79], while others—especially in borderline patients [78]—find effects largely limited to nightmares and hyperarousal. This variability may stem from the depth of traumatic disorganization. In personality disorders, traumatic representations are often fragmented or disorganized, making them less accessible to imagery-based restructuring. Moreover, mental imagery therapy relies on the ability to create, modify, and engage with imagery—skills that may be impaired in borderline profiles with significant dissociations, potentially limiting overall effectiveness.

Functionally, the positive correlation between global sleep improvement (PSQI, ISI, sleep VAS) and quality of life ($p = 0.014$) underscores the importance of addressing sleep disturbances in PTSD treatment given their direct impact on daytime functioning and psychological well-being [84]. However, because of the uncontrolled design, these associations should be interpreted cautiously, as observed changes may also reflect non-specific therapeutic effects, regression to the mean, expectancy effects, or the influence of concurrent psychiatric care. The association between nightmare frequency (NDQ-13) and anxiety intensity measured via VAS suggests VAS may represent a simple, non-intrusive clinical tool for monitoring treatment response without reactivating the patient—an important consideration for future studies.

The lack of change in chronotype after IRT aligns with its nature as a stable biological trait unlikely to shift with short-term psychotherapy. Fatigue, measured by the Pichot scale, also remained unchanged, likely due to multiple contributing factors such as mood, context-specific triggers, or overall sleep quality. Thus, despite reductions in nightmares, persistent symptoms like hypervigilance or anxiety may continue to fuel daytime fatigue.

In our sample, anxiety, depressive symptoms, and emotional lability remained globally stable. These findings, necessarily exploratory, do not allow firm conclusions regarding the impact of IRT on these dimensions, which may either be less directly targeted by the intervention or require a longer time to evolve [62, 77]. This absence of significant effects may reflect IRT's specific focus on traumatic nightmares without addressing broader cognitive and emotional schemas that require more sustained interventions [85]. Notably, the improvements in depressive and anxiety symptoms seen in Sayk's study (2024) [83] were likely driven by multiple factors: an intensive therapeutic setting (inpatient DBT care), group format promoting support and self-efficacy, and repeated imagery training sessions that may have facilitated emotional integration of modified dream content.

No significant improvement was observed in the ALS score, likely reflecting its limitations in capturing the specific emotional dysregulation seen in CPTSD. ALS primarily assesses emotional variability (frequency, intensity) without accounting for key qualitative dimensions such as affective collapse, chronic hyperarousal, or dissociation, reducing its sensitivity in severely dysregulated clinical profiles like CPTSD or BPD [83]. Finally, combining IRT with targeted memory reactivation (TMR) could be a promising avenue, enhancing positive emotional activation in dreams and strengthening the reparative effects of imagery work [86].

Dual coding and personalization of imagery rehearsal therapy: toward mechanistic and clinical optimization

IRT combines verbal rewriting of the dream with guided sensory imagery; this dual coding engages two complementary mnemonic pathways to consolidate a protective script. It is therefore pertinent, clinically and in research, (i) to assess whether its efficacy is moderated by cognitive profile (predominantly "visual" vs. "verbal") and/or individual imagery capacity (assessed with a standardized imagery scale); (ii) to tailor the protocol accordingly (visual reinforcement, auditory supports, co-construction through drawing, storyboards, audio scripts); and (iii) to test imagery training to optimize IRT in low visualizers. The relative contribution of learning techniques (at-home visualization exercises, audio recordings, drawing/comic-book supports, graded imaginal exposure) to consolidating effects and preventing relapse also warrants further investigation.

Several questions remain: (1) which IRT components (psychoeducation, script rewriting, guided imagery, mental rehearsal) contribute most to efficacy, and in what optimal "dose"? (2) In pediatric populations, does efficacy vary according to the nature and degree of representation of integrated protective factors (reassuring figures, means of protection, mastery outcomes)? (3) What is the specific efficacy for amnesic nightmares? In this context, would dream training be effective on somato-affective traces, and through which mediators?

Strengths and limitations of the study

This study fills an important gap by specifically addressing traumatic nightmares in youth with CPTSD exposed to early and repeated traumas—a population that remains understudied. Its strengths lie in the clinical relevance of the topic, a rigorous methodology combining validated tools and qualitative analysis of dream content, and excellent acceptability (0% dropout). The diversity of reported traumas enhances the ecological validity of the results. The absence of dropout in our study, in contrast to Sayk's study (33% among patients with borderline personality disorder) [83], may be explained by the containing, structured, and narrative nature of IRT, which may be particularly compatible with adolescents' psychological functioning (activation of a protective imaginary space, clear structure, short time frame) and may have facilitated therapeutic engagement.

This study has several limitations. The small sample size (39 participants), the absence of a control group, and the short follow-up period limit the strength of the conclusions [87]. In addition, no formal correction for multiple comparisons was applied. As a result, findings—especially those close to the significance threshold—should be interpreted with caution in this exploratory study. Accordingly, the observed improvements should be understood as preliminary associations occurring after treatment rather than as evidence of a specific causal effect of IRT. An additional limitation lies in the developmental and psychometric heterogeneity of the sample, which combined adolescents and young adults assessed with different age-appropriate PTSD instruments. This limits direct comparability across participants and calls for cautious interpretation of PTSD-related outcomes, particularly in the adult subgroup, which remained very small ($n = 5$). In this context, it would be relevant to compare IRT with other brief therapies, such as written exposure therapy (WET), whose protocol is comparable to IRT in that it also consists of five sessions combining psychoeducation and writing, but which has never been tested in adolescents. Although treatment standardization relied on common therapist training, a shared written protocol, and regular supervision meetings, no independent formal rating of treatment fidelity was performed, which should be considered a methodological limitation. At the time of study initiation, no validated French-language tool was available to assess CPTSD in participants under 18; accordingly, diagnosis was based on ICD-11 clinical criteria and supported by the ITQ as a structured complement to clinical evaluation. Future studies should include controlled designs, larger sample sizes, and longitudinal follow-up, as well as post-treatment qualitative analyses of nightmare content in order to better characterize therapeutic transformations (e.g., reduction of violent content, emergence of protective figures) and to anticipate treatment response [61]. Moreover, validating the Nightmare Symptom Inventory (NSI) [88] and the Trauma-Related Nightmare Survey (TRNS) [89] for trauma-related nightmares in youth is an important next step. Finally, the development of digital tools, currently underway within our team, represents a promising avenue to improve accessibility and dissemination of these interventions among young people.

Conclusion

Our study highlights the feasibility, good acceptability, and potential clinical relevance of imagery rehearsal therapy (IRT) as an innovative and developmentally appropriate approach for adolescents and young adults with CPTSD exposed to early and repeated traumatic events. By directly targeting traumatic nightmares and sleep disturbances, IRT may offer a therapeutic entry point associated with improvements in sleep disturbances, nightmare reconditioning, and a greater sense of control over nocturnal and daytime experiences in youth who often exhibit profound learned helplessness resulting from repeated trauma [70]. Its indirect approach to trauma may make it particularly tolerable for patients who are reluctant or unresponsive to exposure-based therapies [76, 81]. The flexibility and brevity of IRT appear compatible with good treatment adherence and acceptability, and were associated in our sample with encouraging short-term clinical changes and no major adverse effects [64]. In addition, the mental imagery skill acquired may be reused autonomously over time. This transferability suggests that the intervention may be adaptable to diverse populations, including those living in contexts of precarity, as illustrated by the

IRT program implemented by the National Association of County and City Health Officials (NACCHO) [90]. Finally, the development of digital and virtual formats—likely to be particularly appealing to youth—may further enhance accessibility and broaden implementation perspectives for adolescents [91]. Overall, our findings suggest that IRT may represent a promising adjunctive intervention for traumatic nightmares in youth with CPTSD, while underscoring the need for controlled studies to clarify its specific effects, long-term outcomes, and place within multimodal care pathways for adolescents exposed to early and repeated violence.

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Author contribution JR, LM, and TZ conducted the literature review and conceived the study. JR, LM, TZ, UH, ER, PB, and CS contributed to protocol development, ethical approval procedures, participant recruitment, and data analysis. JR drafted the first version of the manuscript. All authors critically reviewed and edited the manuscript and approved the final version.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval and consent to participate The study protocol was submitted to the Ethics Committee of the University of Strasbourg, which considered the study to be an interventional clinical audit (reference CE-2023–85). Written informed consent to participate was obtained from all participants. For minors, written consent was obtained from their parents or legal guardians.

Consent for publication Written consent for publication was obtained from all participants. For minors, consent was obtained from their parents or legal guardians. All data were fully anonymized for analysis and reporting.

Conflict of interest The authors declare no competing interests.

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